

WEEK - 4

Time-Series Visualization & Financial Dashboard

Introduction :

In this project, Shopify stock market data was analyzed using Power BI. The dataset contains stock-related information such as Open Price, High Price, Low Price, Close Price, and Trading Volume.

The main focus of this week was to study trading volume using moving averages and an MA Crossover Signal. Different visualizations, slicers, a Metric Selector, and a custom tooltip were created to make the dashboard interactive and easier to analyze.

Objectives :

The main objectives of this project are:

- To analyze Shopify stock market data and study changes in stock price and trading volume over time.
- To calculate 20-day and 50-day moving averages for trading volume.
- To compare the moving averages using an MA Crossover Signal.
- To highlight important changes in trading volume using Volume Color.
- To create suitable visualizations for analyzing stock price, trading volume, moving averages, and daily returns.
- To use slicers and a Metric Selector for interactive data analysis.
- To create a custom tooltip for viewing detailed stock information.
- To present the analysis through an interactive and informative Power BI dashboard.

Measures and Calculations :

The following calculations were created as part of the Week 4 analysis.

Volume 20 SMA :

```
Volume 20 SMA =  
VAR CurrentDate = MAX('Date Table'[Date])  
VAR PeriodDays = 20  
VAR DateInPeriod =  
    DATESINPERIOD(  
        'Date Table'[Date],  
        CurrentDate,  
        -PeriodDays,  
        DAY  
    )  
RETURN  
    CALCULATE(  
        AVERAGE('SHOP_2015-05-21_2025-03-16'[volume]),  
        DateInPeriod  
    )
```

This calculation was created to find the average trading volume over a 20-day period and observe the shorter-term volume trend.

Volume 50 SMA :

```
Volume 50 SMA =  
VAR CurrentDate = MAX('Date Table'[Date])  
VAR PeriodDays = 50  
VAR DateInPeriod =  
    DATESINPERIOD(  
        'Date Table'[Date],  
        CurrentDate,  
        -PeriodDays,  
        DAY  
    )  
RETURN  
    CALCULATE(  
        AVERAGE('SHOP_2015-05-21_2025-03-16'[volume]),  
        DateInPeriod  
    )
```

This calculation was created to find the average trading volume over a 50-day period and provide a longer-term view of the volume trend.

Volume Color :

```
Volume Color =  
IF([Trading Volume] > [Volume 20 SMA] * 1.5, "#FF4D4D", "#26A69A")
```

This calculation was used to highlight higher trading volume in the dashboard and make changes in volume easier to notice.

MA Crossover Signal :

```
MA Crossover Signal = IF( ISBLANK([Volume 20 SMA]) || ISBLANK([Volume  
50 SMA]), BLANK(), IF([Volume 20 SMA] > [Volume 50 SMA], "Bullish (20 > 50)",  
"Bearish (20 < 50)") )
```

This calculation compares the 20-day and 50-day moving averages and provides a signal based on their relationship.

Visualizations :

The dashboard was divided into different pages so that the stock data could be viewed from different perspectives.

Stock Analysis Visuals

Line and Stacked Column Chart :

This chart was created to study the relationship between the stock's closing price and its trading-volume moving averages over time. It helps in comparing the short-term and long-term volume trends along with the closing price.

Price Analysis Chart :

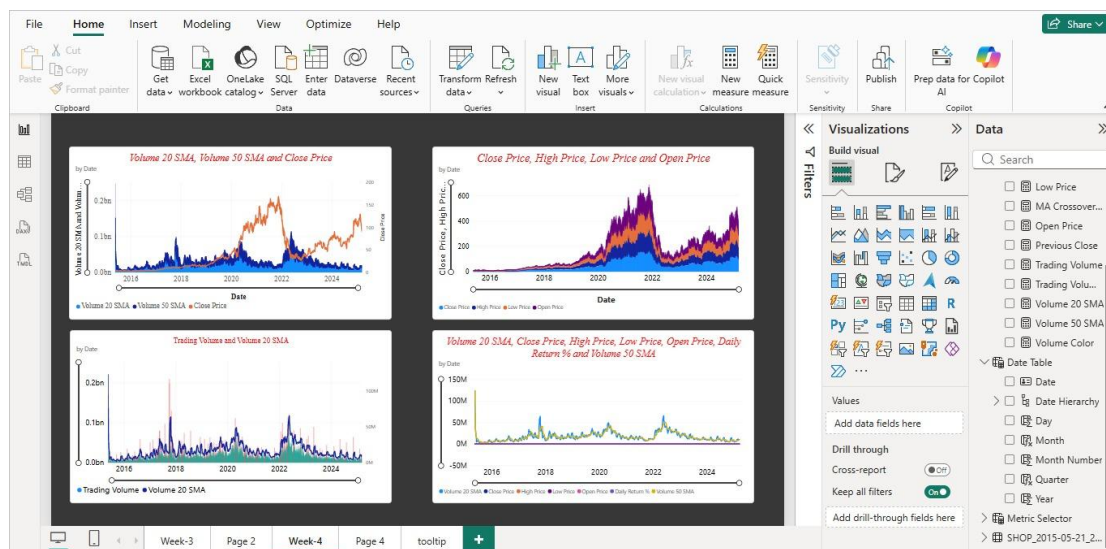
This chart was created to view the different stock price values together. It helps in understanding how the stock price changed over the selected period.

Line and Stacked Column Chart – Trading Volume Analysis :

This chart was created to compare actual trading volume with its 20-day moving average. **Volume Color** was applied to the Trading Volume columns to make higher volume activity easier to identify.

Line Chart :

This chart was created to observe the movement of the different stock measures and moving averages together over time. It also includes Daily Return % to provide an additional view of daily price movement.



Slicers and Interactive Analysis

Slicers were added to make the dashboard easier to explore and allow the required time period or price measure to be selected.

Date Slicer :

The Date Slicer was added to select a particular period from the available stock data. This helps in focusing the dashboard analysis on the required dates.

Relative Date Slicer :

The Relative Date Slicer was added to provide another date-based filtering option. It helps in working with the data based on a relative time period.

Metric Selector :

The Metric Selector was created using a **Field Parameter**. It allows the user to select the required price measure such as Close Price, High Price, Low Price, or Open Price without creating separate visuals for each measure.

```
Metric Selector = {  
    ("Close Price", NAMEOF('Measures Table'[Close Price]), 0),  
    ("High Price", NAMEOF('Measures Table'[High Price]), 1),  
    ("Low Price", NAMEOF('Measures Table'[Low Price]), 2),  
    ("Open Price", NAMEOF('Measures Table'[Open Price]), 3)  
}
```

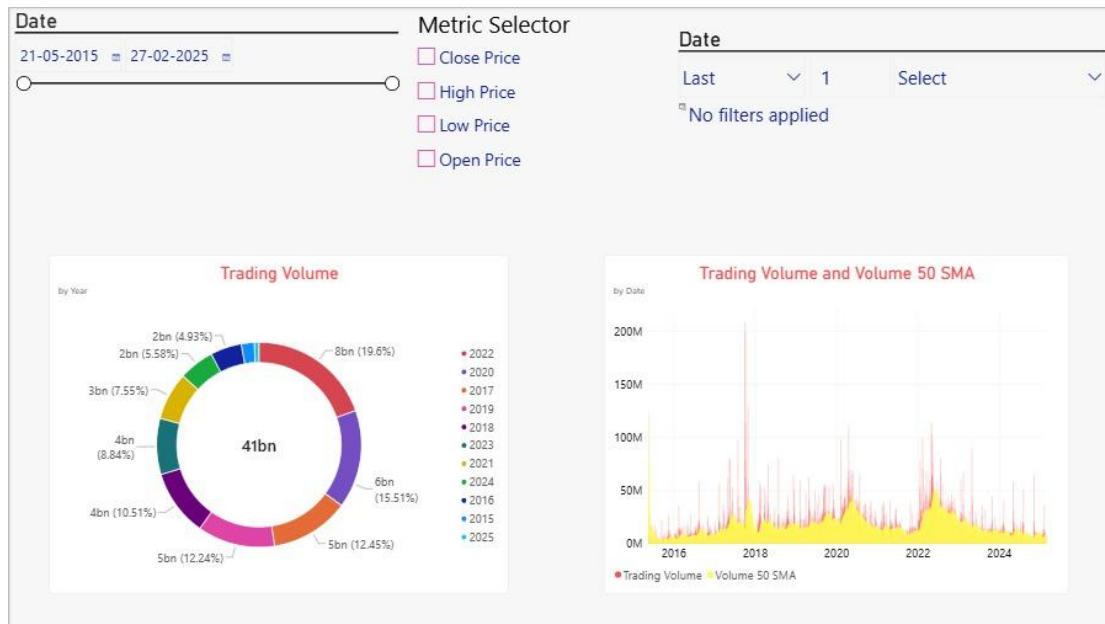
Trading Volume Analysis :

✧ Donut Chart :

The Donut Chart was created to compare the contribution of trading volume across different years. It provides a simple overall view of how the trading volume is distributed year-wise.

✧ Clustered Column Chart :

The Clustered Column Chart was created to compare Trading Volume with the 50-day moving average. It helps in observing how the actual volume differs from the longer-term volume trend.



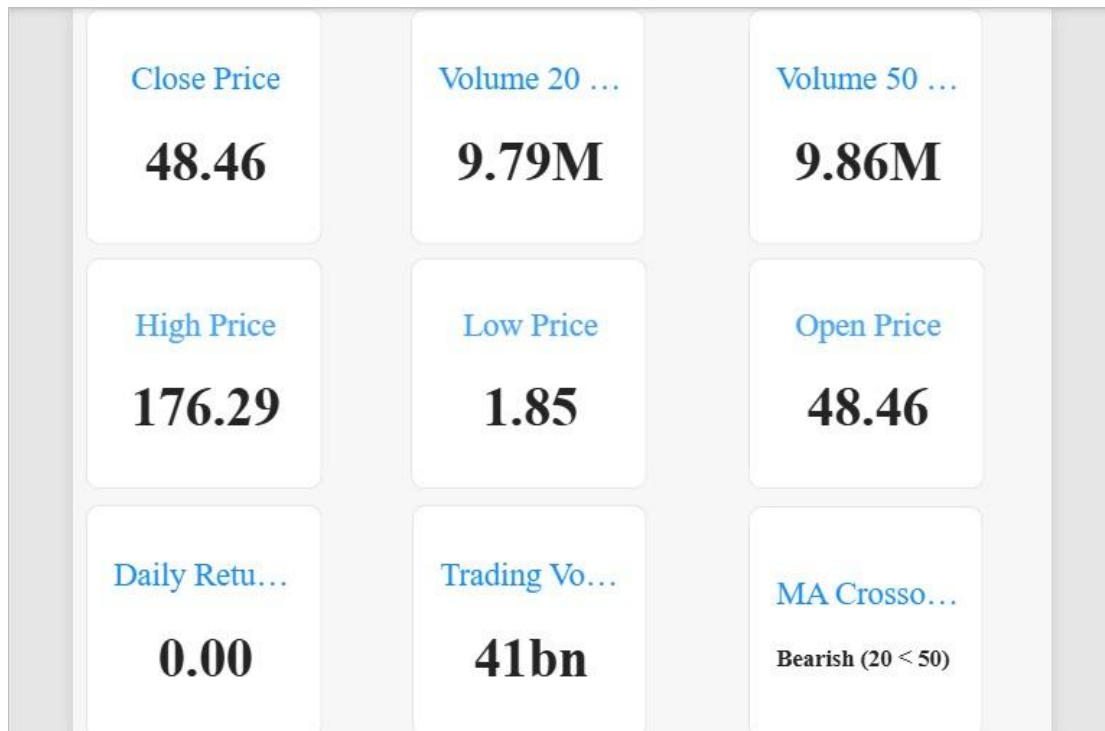
Custom Tooltip :

A custom tooltip was created to provide additional information when interacting with the dashboard visuals. Instead of placing all the details directly on the main dashboard, the tooltip provides the required information when the user views a particular data point.

The tooltip contains the following cards:

1. Close Price
2. High Price
3. Low Price
4. Open Price
5. Daily Return %
6. Trading Volume
7. Volume 20 SMA
8. Volume 50 SMA
9. MA Crossover Signal

These cards provide a detailed view of the selected stock data while keeping the main dashboard clean.



Key Insights :

The dashboard provides a clear way to study stock price and trading volume data over time. The 20-day and 50-day moving averages help in viewing trading volume trends over different time periods.

The MA Crossover Signal provides a simple comparison between the two moving averages, while Volume Color helps to highlight higher trading volume activity. The price-related visuals allow Open, High, Low, and Close values to be viewed together.

The Date Slicer, Relative Date Slicer, and Metric Selector make the dashboard interactive. The Donut Chart provides a year-wise view of trading volume, while the Clustered Column Chart helps compare trading volume with its 50-day moving average. The custom tooltip provides additional details without making the main dashboard crowded.

Conclusion :

This project focused on analyzing Shopify stock market data using Power BI. The main calculations included Volume 20 SMA, Volume 50 SMA, Volume Color, and MA Crossover Signal for studying trading volume and its moving averages.

Different visualizations were created along with date-based slicers, a Metric Selector, and a custom tooltip. These features helped to organize the stock information clearly and provided an interactive way to explore price and trading volume data.

Overall, the completed dashboard presents the stock market data in a clear, structured, and interactive format.